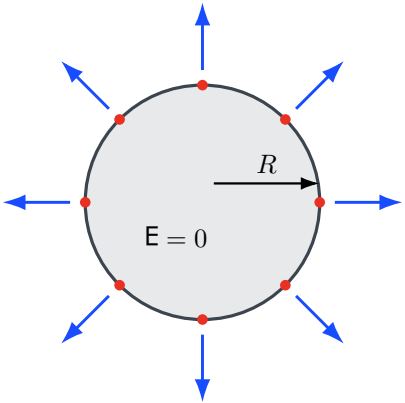
孤立带电导体球：外部等效点电荷，内部等势且电场为零



$$r \geq R: \quad \mathsf{E} = \frac{kQ}{r^2}\mathsf{e}_r, \quad \varphi = \frac{kQ}{r}$$

$$r \leq R: \quad \varphi = kQ/R, \quad \sigma = Q/(4\pi R^2), \quad C = 4\pi\varepsilon_0 R$$

$$U = \frac{Q^2}{8\pi\varepsilon_0 R} = \frac{1}{2}QV$$